ICE 11.1

Aman Gupta • Apr 10, 2025 (Edited Apr 27, 2025)

8 points

1. (2 marks) Refer to the complete description of the Kerberos v4 protocol. How is it that the Authentication server is able to authenticate client C without client C's password being sent over the network?

- Password is not required since both client C and AS have the client password
- Password is indeed sent by client C over the network, except that it is included but encrypted in the message 1
- **The client which posseses the password K_C will be able to decrypt message 2, and thus obtain Ticket_tgs (correct)**
- None of the above

2. (2 marks) Refer to Kerberos protocol in Question 1, above. What makes it possible for client C to gain access to a server V over and over again without having to share its password with AS, TGS or the server V?

- Not quite. Client C has to reinitiate dialogue with AS, TGS and then access server V
- Not quite. Client C has to reinitiate dialogue with TGS and then access server V
- **Whenever client C wishes to access server V, message 5 is re-sent by client C (correct)**
- None of the above

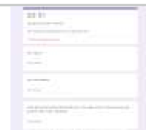
3. (2 marks) Refer to Kerberos protocol given Question 1, above. What makes it possible for client C to authenticate a server V?

- **Server V is authenticated by TGS, which shares a session key K_(C, V) with both client C and server V (correct)**
- Server V sends its credentials to client C
- TGS authenticates server V on behalf of client C
- None of the above

4. (2 marks) What aspect of the Kerberos protocol in Question 1 ensures that no message sent by a client other than client C is entertained by the server V?

- Client C must send its own password to server V along with Ticket_V
- Client C must send the Authenticator_C to server V
- **Client C must send the Authenticator_C together with the Ticket_V (correct)**
- None of the above

[ICE 11.1](#)
Google Forms



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